Performance Evaluation of Basic Education Informatization Based on Improved BP Algorithm

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***Abstract*—In recent years, with the development of educational informatization, the problem of educational informatization has become increasingly prominent, which makes educators have to pay attention to the performance evaluation of educational informatization. The education work focuses more on the integration of various forms of education. In order to solve the limitations of basic education informatization, this paper analyzes the BP algorithm function and basic education informatization performance evaluation index, designs and discusses the network parameter setting and performance evaluation index data collection of applied research, and analyzes basic education information. The BP calculation flow chart of the performance evaluation is designed, and finally the experimental data analysis is carried out for the expected evaluation value, actual evaluation value and training evaluation value of the designed algorithm in the performance evaluation of education. The experimental data shows that the expected evaluation value and the training evaluation value are analyzed. The difference of the actual evaluation value fluctuates in the range of 0.1 to 0.5 on average, thus verifying the feasibility of applying the BP algorithm in the performance evaluation of various schools in and out of elementary education.**

Keywords—BP algorithm, basic education informatization, performance evaluation, basic education

I. Introduction

In the new wave of informatization, the role and influence of information technology on society has attracted the attention of many people. Therefore, the reform and adjustment of education informatization has become an inevitable trend.

Nowadays, more and more scholars conduct research on the practical application of improved BP algorithm in the performance evaluation of basic education informatization, and have achieved certain research results. The purpose of Lee C S research is to identify and design the links between modern social development and higher education systems through information and communication technologies. The novelty of the Lee C S study is the identification of an approach to information awareness in higher education systems as a stand-alone approach. Lee C S is based on applied theoretical research methods. The methodological basis of Lee CS's research is the conceptual approach and the appeal to the origin of modern educational informatization. In the field of research carried out, according to the analysis of educational activities and social philosophy of education, the close causal relationship between the development of modern social informatization and the development of education has been obtained [1]. Farooq proposed that the performance evaluation of educators aims to play a role in the quality and professional development of school education, and then improve the learning outcomes of students. Data were collected from 20 schools in a medium provincial and urban area that were assessed over a semester. The performance of 20 schools is also described so that gaps in the quality of school education can be examined. The results show that most schools are rated within the standard performance range on the quality assessment items [2]. Rodrigues MW proposed a new SER-based school performance evaluation method, which can play a role in the performance evaluation of school education. The method consists of three parts: First, the expectation and idea of the school's service quality are investigated and analyzed through questionnaires. In the second part, the corresponding weight of each dimension of school education quality evaluation is determined by cluster analysis. In the third part, the PSIS method is used to rank each dimension of service quality of school education. The method proposed by Rodrigues MW has certain reference significance for the performance evaluation of school education service quality [3]. Although the existing performance evaluation research has been very successful, there are still many problems in education performance evaluation research on the basis of improving the BP algorithm.

Therefore, in order to solve the problem of performance evaluation in education, this paper uses BP algorithm to design and application. First, the function of this algorithm and the specific indicators of performance evaluation in education are introduced, and then the network parameter setting and performance of model design and experiment are discussed. The evaluation index data collection, and finally design the model structure of the improved BP algorithm in the basic education informatization experimental application, and through the experiments on the specific application model. Finally, the feasibility of the improved BP algorithm is demonstrated.

II. Performance Evaluation of Basic Education Informatization Based on Improved BP Algorithm

A. BP Learning Algorithm

(1) Enter a sample of performance evaluation items

The performance evaluation item sample of BP algorithm is divided into five parts into input vector $S_v^l = \left[s_1^l, s_2^l, \ldots s_x^j\right]$ and the corresponding performance

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evaluation expected output vector $d_v^l = \left[d_1^l, d_2^l, .., d_x^l\right]$, d represents the first evaluation mode.

(2) The formula for calculating the input activation value g_v^l and response output value s_v^l of the hidden layer performance evaluation project is as follows:

$$g_v^l = \sum_{u=1}^{r} M_{uv} s_v^l - \varpi_v \tag{1}$$

Among them, $v = 1,2\int$. The response output value of each performance evaluation of the hidden layer is $s_v^l = r(f_f^l)$, and the function r is the activation function of the performance evaluation of the hidden layer.

(3) Calculate the activation value g_v^l and the response value s_v^l of the hidden layer performance evaluation item.

The calculation formula of the input activation value of each evaluation item in the output layer of the evaluation quality output value is:

$$g_v^l = \sum_{u=1}^{r} M_{uv} s_v^l - \varpi_v \tag{2}$$

Among them, $v = 1,2\int$. The response output value of each performance evaluation of the hidden layer is $s_v^l = r(f_f^l)$, and the function r is the activation function of the performance evaluation of the hidden layer [4].

(4)Calculate the sum of squares error A of the network

In the B-P algorithm, the sum of squares error A of the network can be used to judge whether the actual performance evaluation output of the network output layer is consistent with the expected performance evaluation output. The calculation formula is as follows:

$$G = \sum_{l-1}^{n} G_k \tag{3}$$

Where G_k is the error of actual performance evaluation and expected performance evaluation [5].

B. Performance Evaluation Indicators of Basic Education Informatization

The research on the performance evaluation of basic education informationization is essentially based on the whole evaluated school, so as to evaluate the individual. Each school is a decision-making unit, and the input and output of each school form an input-output system. Formulating evaluation indexes for school input and output is fundamental to school performance assessment [6].

(1) Input indicators

The input indicators of school performance include information management and input (A), information infrastructure construction (B), information teaching resources construction (C), information talent team construction (D), teacher information literacy construction (E) and other input indicators [7].

1) Information management and input include: organizational planning (A1), system construction (A2), management and maintenance (A3), funds input (A4) and other input indicators

2) Information infrastructure construction includes input indicators such as teaching equipment construction (B1) and campus network construction (B2) [8].

3) The construction of informationized teaching resources includes the quantity of teaching resources (C1) input index [9].

4) Information talent team construction includes: information full-time teachers (D1) input index.

5) Teacher information literacy construction includes: teacher information technology training (E1) input indicators [10].

(2) Output indicators

The output indicators of school performance include: information infrastructure construction (B), information talent team (C), teacher information literacy construction (B) [11].

1) Information infrastructure construction mainly refers to: campus network construction (B3) output index.

2) The construction of information teaching resources includes: self-built teaching resources (C2) output index.

3) The construction of information talent team includes: information full-time teachers (D2) and information administrators (D3) and other output indicators.

4) Teachers' information literacy construction includes: teachers' ability to understand information technology (E2), teachers' ability to operate information technology (E3), teachers' ability to carry out instructional design in information environment (E4) and other output indicators.

Students' informatization ability includes knowledge and emotion (F1), basic operation ability (F2), information acquisition, analysis and processing ability (F3) and the ability to use information technology to promote learning (F4) and other output indicators.

4) Teachers' information literacy construction includes output indicators such as teachers' ability to understand information technology (E2), teachers' ability to operate information technology (E3), and teachers' ability to design instruction in an information environment (E4).

5) Students' informatization ability includes: knowledge and emotion (F1), basic operation ability (F2), information acquisition, analysis, processing ability (F3) and the ability to use information technology to promote learning (F4) and other output indicators.

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III. Investigation and Research on Performance Evaluation of Basic Education Informatization Based on Improved BP Algorithm

A. Network Parameter Setting of Basic Education Informatization Performance Evaluation Based on Improved BP Algorithm

The algorithm constructed in this paper consists of three layers: input layer, hidden layer, and output layer. The number of nodes in each layer is 17, 10, and 1, respectively. When training the BP neural network, the training will stop as long as one of the following two conditions are met. The two conditions are: ① the set test data is reached; ② the set training accuracy target is reached (this paper is set to le-3) . In addition, the setting of learning rate should be small rather than large, so this paper sets the rate of performance evaluation to 0.05.

B. Evaluation Data Collection of Basic Education Informatization Performance Evaluation Based on Improved BP Algorithm

We take the first five groups of basic education informatization performance evaluation indicators as input, namely (informatization management and investment, informatization infrastructure construction, student informatization ability, informatization talent team construction, teacher information literacy construction), and the corresponding basic education Education informatization performance evaluation as output. And use the BP algorithm function that comes with MATLAB to normalize these data. The compiled datasets are shown in Table I:

TABLE I. Basic education informatization performance index evaluation data setting

serial number	Comprehensive evaluation score	Information infrastructure construction	Information technology talent team building	Teachers' information literacy construction
1	3.56	3.14	3.345	3.215
2	2.15	2.65	3.126	3.014
3	3.016	3.18	3.257	3.127
4	3.217	3.214	2.698	3.241

IV. Research and Application of Basic Education Informatization Performance Evaluation Based on Improved BP Algorithm

A. Design of Basic Education Informatization Performance Evaluation Model Based on Improved BP Algorithm

Before the construction of the BP neural network, the key point is to determine the network's input layer and output layer, that is to say, the performance evaluation of informatization of education investment, infrastructure construction, teachers, students, talents construction quality construction objects one by one to determine each evaluation index, and as input values of BP neural network system, the fuzzy comprehensive evaluation Is the output value of the system.

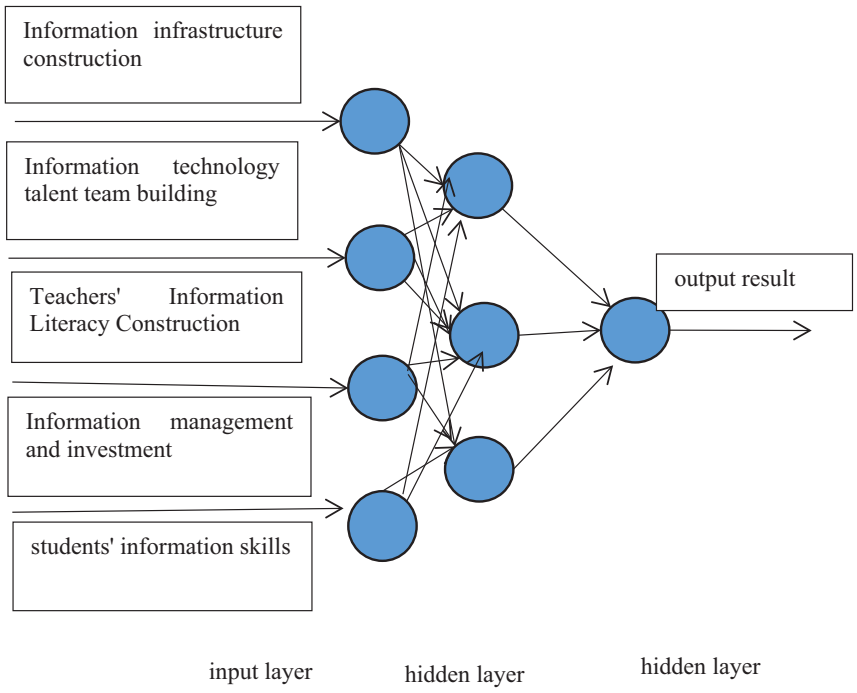


Fig. 1. Evaluation network structure model design

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As shown in Figure 1, taking information management and investment, information infrastructure construction, students' information ability, information talent team building, teachers' information literacy construction and original score as the input of a neural network, it can be established as shown in Figure 1 show the network structure. Among them, the five input layers are information management and investment, information infrastructure construction, students' information ability, information talent team construction, and teachers' information literacy construction. There are 12 nodes in the original hidden layer. The output layer has 1 node, which is the y evaluation value, and the value range is [1-5].

B. Application of Basic EducationInformatization Performance Evaluation Model Based on Improved BP Algorithm

According to the BP neural network constructed in this paper, the performance evaluation results are compared using the expected evaluation value calculated by the BP algorithm, where A represents information management and investment, B represents information infrastructure construction, C represents students' information ability, D represents the construction of informatization talent team, and E represents the construction of teachers' information literacy. The specific data is shown in Figure 2:



Fig. 2. Comparison of expected output value, actual output value, and test output value

According to the BP algorithm of the performance evaluation of the project classification, evaluation method based on BP neural network to the elementary education informationization in the performance management and investment, information infrastructure construction, evaluation are classified and the student's ability of informatization evaluation of desired output and the error between actual output values and the training evaluation value is very small, It shows that BP algorithm is effective in evaluating the performance of basic education informatization.

V. Conclusion

Therefore, in order to enrich the application research between BP algorithm and the performance evaluation of basic education informatization, this paper firstly summarizes the algorithm and the performance evaluation of basic education informatization in this paper, and investigates and designs the network parameter setting and performance evaluation index data collection of the experiment in this paper. Secondly, it designs and analyzes the combination of BP algorithm and basic education informatization and the practice model. Finally, it compares and analyzes the experimental data for the performance evaluation value of the algorithm in specific applications. The final experimental results verify the use value of BP algorithm in performance evaluation.

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